

Discrete Random Variables

Department of Computer Science

University of Botswana

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Learning Objectives

By the end of this lecture, you should be able to:

- define a random variable and distinguish discrete from continuous random variables;
- construct a PMF from a sample space;
- check whether a function is a valid PMF;
- compute probabilities, expectation, variance and standard deviation;
- recognize common discrete distributions: Bernoulli, uniform, binomial and geometric;
- solve exam-style problems involving discrete random variables.

Motivation

Many experiments have outcomes that are not numbers.

Examples

- Tossing coins: outcome can be HHT, TTH, etc.
- Rolling two dice: outcome can be $(3, 5)$.
- Observing a system: outcome can be success or failure.

Key idea

A random variable assigns a number to each outcome so that we can calculate probabilities using numbers.

Random Variables: The Idea

Definition

A **random variable** is a function that assigns a real number to every outcome in the sample space.

$$X : \Omega \longrightarrow \mathbb{R}$$

Example: two dice

Let the outcome be (d_1, d_2).

- $X = d_1 + d_2$ is the sum of the two dice.
- $Y = |d_1 - d_2|$ is the absolute difference.
- $Z = \max(d_1, d_2)$ is the maximum.

Notation

- Use capital letters for random variables: X, Y, Z.
- Use lowercase letters for values: x, y, z.
- The statement $X = x$ is an event.

Example

If X is the sum of two dice, then

$$\{X = 7\} = \{(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)\}.$$

Therefore,

$$\mathbb{P}(X = 7) = \frac{6}{36} = \frac{1}{6}.$$

Discrete and Continuous Random Variables

Discrete random variable

A random variable is **discrete** if it takes values in a finite or countably infinite set.

Continuous random variable

A random variable is **continuous** if it can take values in an interval of real numbers.

Classify

- Number of heads in 3 coin tosses: discrete.
- Sum of two dice: discrete.
- Time until the next request arrives at a server: continuous.
- CPU processing time for a request: continuous.

From Random Variables to Events

When we set a random variable equal to a value, we get an event.

Random variable	Event	Probability
$X =$ sum of two dice	$X = 7$	$\mathbb{P}(X = 7) = 1/6$
$N =$ arrivals in one hour	$N > 10$	$\mathbb{P}(N > 10)$
$Y =$ heads in 3 tosses	$Y = 2$	$\mathbb{P}(Y = 2) = 3/8$

Important

The probability rules for events still apply to random variables.

Example: Three Fair Coins

We flip three fair coins and define

$$Y = \text{total number of heads observed.}$$

Possible values:

$$Y \in \{0, 1, 2, 3\}.$$

Note

$Y = 4$ is impossible because only three coins are tossed.

Sample Space for Three Coin Tosses

The sample space has $2^3 = 8$ equally likely outcomes:

$$\Omega = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}.$$

Each outcome has probability

$$\frac{1}{8}.$$

Counting idea

To find $\mathbb{P}(Y = k)$, count the outcomes with exactly k heads.

PMF of Y: Number of Heads in Three Tosses

y	Outcomes	$p_Y(y) = \mathbb{P}(Y = y)$
0	TTT	$1/8$
1	HTT, THT, TTH	$3/8$
2	HHT, HTH, THH	$3/8$
3	HHH	$1/8$

$$\mathbb{P}(Y = k) = \frac{\binom{3}{k}}{8}, \quad k = 0, 1, 2, 3.$$

Probability Mass Function

Definition

The **probability mass function** (PMF) of a discrete random variable X assigns probabilities to possible values of X.

$$p_X(x) = \mathbb{P}(X = x)$$

PMF properties

A function $p_X(x)$ is a valid PMF if

$$p_X(x) \geq 0 \quad \text{for all } x$$

and

$$\sum_x p_X(x) = 1.$$

Checking Whether a PMF is Valid

Suppose

$$p_X(0) = 0.2, \quad p_X(1) = 0.5, \quad p_X(2) = 0.3.$$

Check 1: non-negative probabilities

All probabilities are non-negative.

Check 2: probabilities sum to 1

$$0.2 + 0.5 + 0.3 = 1.$$

Conclusion

This is a valid PMF.

Example: Rolling One Fair Die

Let $X =$ outcome of rolling a fair six-sided die.

$$X \in \{1, 2, 3, 4, 5, 6\}.$$

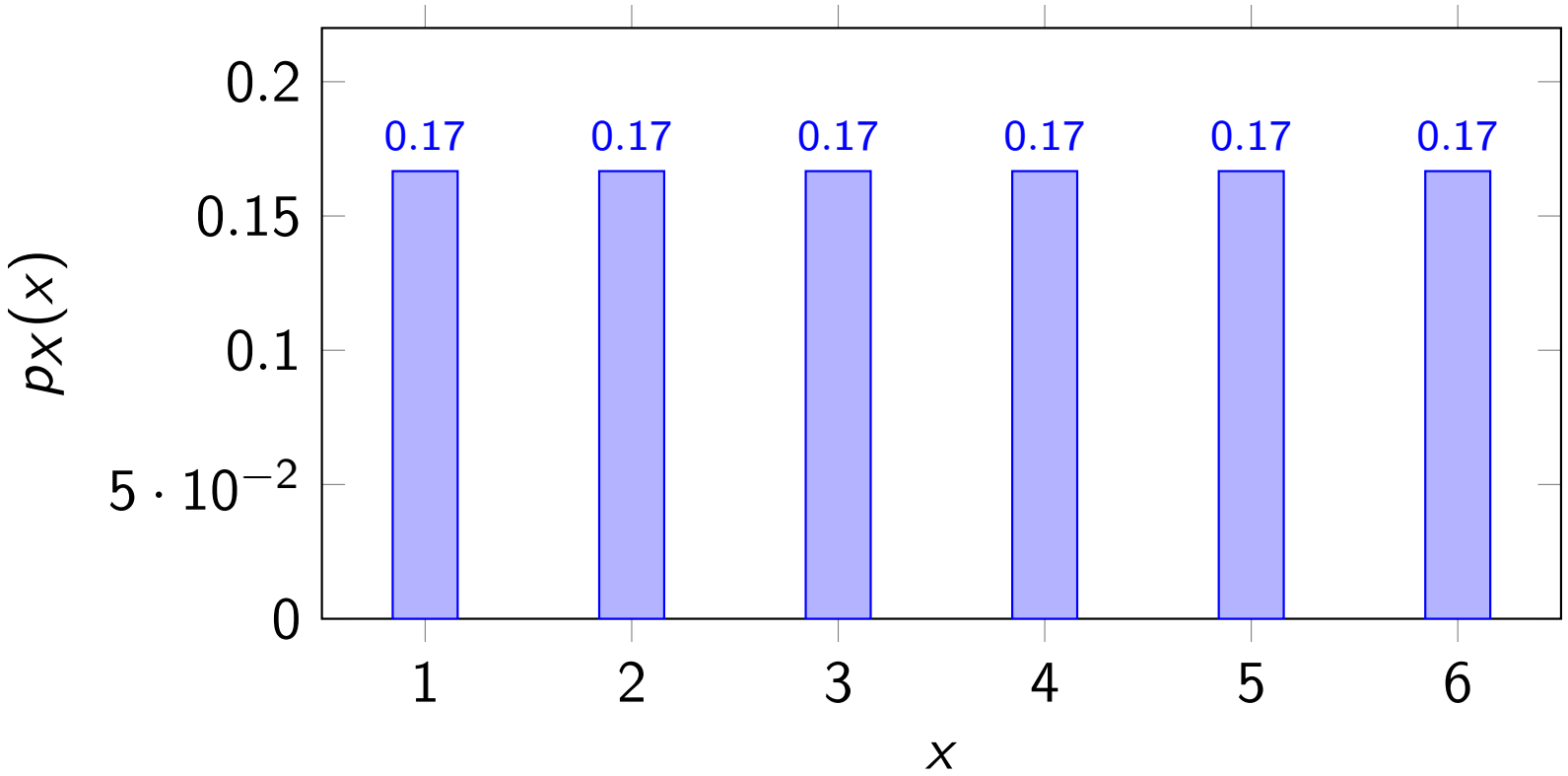
Since all outcomes are equally likely,

$$p_X(x) = \mathbb{P}(X = x) = \frac{1}{6}, \quad x = 1, 2, 3, 4, 5, 6.$$

Otherwise,

$$p_X(x) = 0.$$

PMF Bar Chart: Fair Die



Example: Sum of Two Fair Dice

Let

$$X = d_1 + d_2.$$

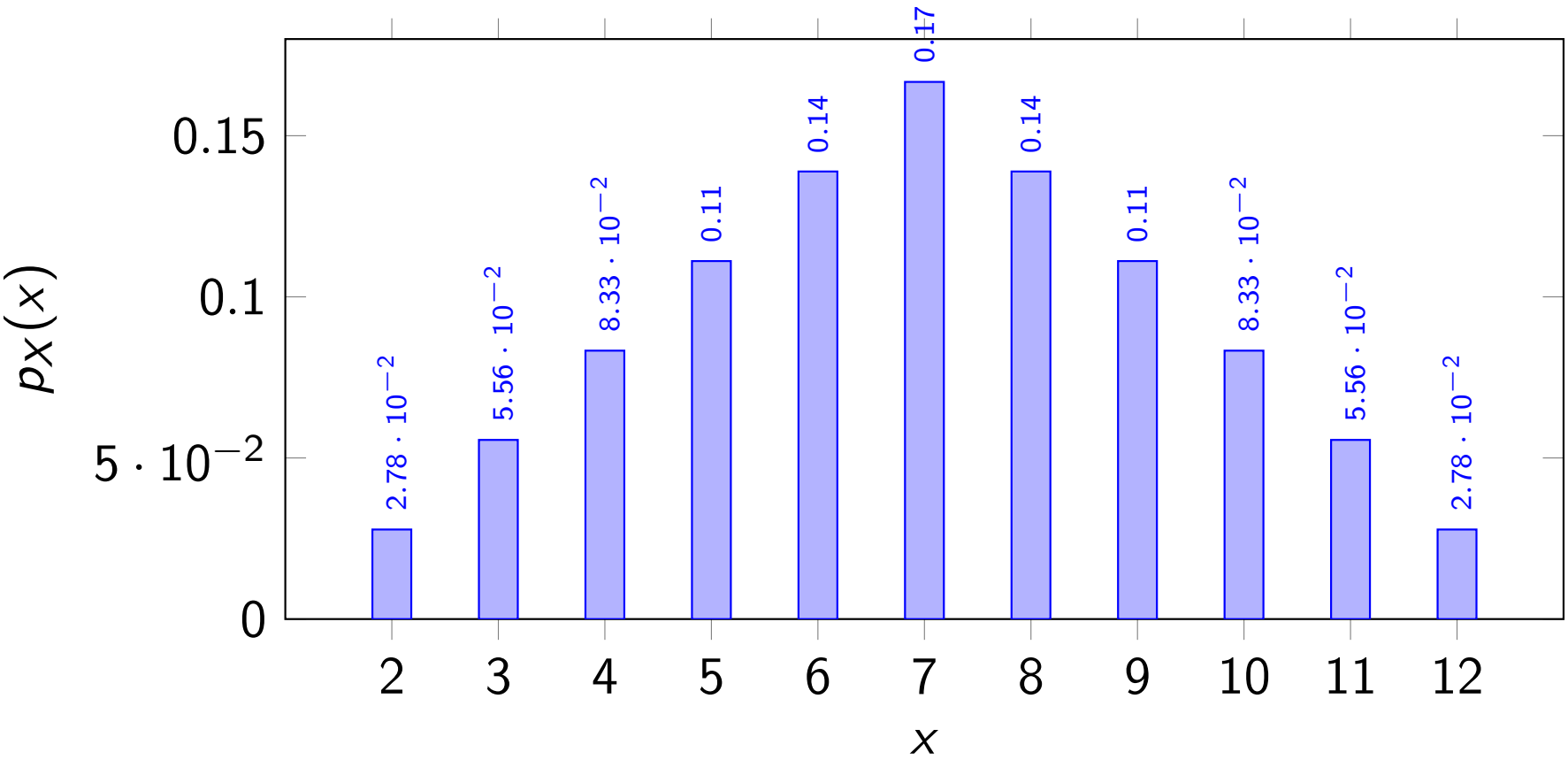
There are 36 equally likely outcomes.

Counting formula

$$p_X(x) = \frac{\#\{(d_1, d_2) : d_1 + d_2 = x\}}{36}, \quad x = 2, 3, \ldots, 12.$$

x	2	3	4	5	6	7	8	9	10	11	12
Ways	1	2	3	4	5	6	5	4	3	2	1

PMF of the Sum of Two Fair Dice



Cumulative Distribution Function

Definition

The cumulative distribution function (CDF) of X is

$$F_X(x) = \mathbb{P}(X \leq x).$$

For a discrete random variable

$$F_X(x) = \sum_{t \leq x} p_X(t).$$

Example

If X is the outcome of a fair die, then

$$F_X(3) = \mathbb{P}(X \leq 3) = \frac{3}{6} = \frac{1}{2}.$$

Expected Value

Definition

The **expected value** or **mean** of a discrete random variable X is

$$\mathbb{E}[X] = \sum_x x p_X(x).$$

- It is a weighted average of the possible values of X.
- Values with higher probability contribute more to the mean.

Example: Expected Value of a Fair Die

Let X be the outcome of a fair die.

$$\mathbb{E}[X] = \sum_{x=1}^{6} x\mathbb{P}(X = x).$$

Substitute the PMF:

$$\mathbb{E}[X] = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6}.$$

$$\mathbb{E}[X] = \frac{1+2+3+4+5+6}{6} = \frac{21}{6} = 3.5.$$

Example: Expected Heads in Three Tosses

For $Y =$ number of heads in three fair coin tosses:

y	$p_Y(y)$
0	$1/8$
1	$3/8$
2	$3/8$
3	$1/8$

$$\mathbb{E}[Y] = 0 \cdot \frac{1}{8} + 1 \cdot \frac{3}{8} + 2 \cdot \frac{3}{8} + 3 \cdot \frac{1}{8} = \frac{12}{8} = 1.5.$$

Expected Value Rule

If $Y = g(X)$, then

$$\mathbb{E}[g(X)] = \sum_x g(x) p_X(x).$$

Example

Let X be the outcome of a fair die. Find $\mathbb{E}[X^2]$.

$$\mathbb{E}[X^2] = \sum_{x=1}^{6} x^2 \frac{1}{6} = \frac{1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2}{6} = \frac{91}{6}.$$

Variance

Definition

The variance of X measures how spread out the values of X are around the mean.

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2], \quad \mu = \mathbb{E}[X].$$

Useful computational formula

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

Standard deviation

$$\sigma_X = \sqrt{\text{Var}(X)}.$$

Example: Variance of a Fair Die

From the previous example,

$$\mathbb{E}[X] = \frac{7}{2}, \quad \mathbb{E}[X^2] = \frac{91}{6}.$$

Therefore,

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

Substitute:

$$\mathrm{Var}(X) = \frac{91}{6} - \left(\frac{7}{2}\right)^2 = \frac{91}{6} - \frac{49}{4} = \frac{35}{12}.$$

$$\sigma_X = \sqrt{\frac{35}{12}} \approx 1.708.$$

Properties of Expectation and Variance

Linearity of expectation

For constants a, b,

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b.$$

For random variables X and Y,

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y].$$

Variance of a linear transformation

$$\mathrm{Var}(aX + b) = a^2 \,\mathrm{Var}(X).$$

Bernoulli Random Variable

Definition

A Bernoulli random variable models one trial with two outcomes: success or failure.

$$X = \begin{cases} 1, & \text{with probability } p, \\ 0, & \text{with probability } 1-p. \end{cases}$$

$$\mathbb{P}(X=1) = p, \quad \mathbb{P}(X=0) = 1-p.$$

$$\mathbb{E}[X] = p, \quad \mathrm{Var}(X) = p(1-p).$$

Example: Bernoulli Distribution

A system request succeeds with probability 0.9. Let

$$X = \begin{cases} 1, & \text{request succeeds,} \\ 0, & \text{request fails.} \end{cases}$$

Then $X \sim \text{Bernoulli}(0.9)$.

$$\mathbb{P}(X = 1) = 0.9, \quad \mathbb{P}(X = 0) = 0.1.$$

$$\mathbb{E}[X] = 0.9,$$

$$\text{Var}(X) = 0.9(1 - 0.9) = 0.09.$$

Discrete Uniform Random Variable

Definition

X is discrete uniform on $\{a, a+1, \ldots, b\}$ if all values are equally likely.

$$p_X(x) = \frac{1}{b-a+1}, \quad x = a, a+1, \ldots, b.$$

Fair die

A fair die is uniform on $\{1, 2, 3, 4, 5, 6\}$.

$$p_X(x) = \frac{1}{6}.$$

Binomial Random Variable

Definition

$X \sim \text{Binomial}(n, p)$ counts the number of successes in n independent Bernoulli trials, each with success probability p.

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \ldots, n.$$

$$\mathbb{E}[X] = np,$$

$$\text{Var}(X) = np(1-p).$$

Example: Binomial Distribution

A quiz has 5 true/false questions. A student guesses randomly. Let $X =$ number of correct answers.

$$X \sim \text{Binomial}\left(5, \frac{1}{2}\right).$$

Find $\mathbb{P}(X = 3)$.

$$\mathbb{P}(X = 3) = \binom{5}{3}\left(\frac{1}{2}\right)^3\left(\frac{1}{2}\right)^2.$$

$$\mathbb{P}(X = 3) = 10\left(\frac{1}{2}\right)^5 = \frac{10}{32} = 0.3125.$$

Geometric Random Variable

Definition

$X \sim \text{Geometric}(p)$ counts the trial number on which the first success occurs.

$$\mathbb{P}(X = k) = (1-p)^{k-1}p, \quad k = 1, 2, 3, \ldots$$

$$\mathbb{E}[X] = \frac{1}{p}.$$

Interpretation

If success probability is small, we expect to wait longer for the first success.

Example: Geometric Distribution

A network packet is successfully transmitted with probability 0.8 on each attempt. Let $X =$ attempt number on which the first success occurs.

$$X \sim \text{Geometric}(0.8).$$

Find $\mathbb{P}(X = 3)$.

$$\mathbb{P}(X = 3) = (1 - 0.8)^{3-1}(0.8).$$

$$\mathbb{P}(X = 3) = 0.2^2 \times 0.8 = 0.032.$$

Expected number of attempts:

$$\mathbb{E}[X] = \frac{1}{0.8} = 1.25.$$

Joint PMF

When two discrete random variables are defined on the same experiment, their joint PMF is

$$p_{X,Y}(x,y) = \mathbb{P}(X = x, Y = y).$$

Marginal PMFs

$$p_X(x) = \sum_y p_{X,Y}(x,y), \quad p_Y(y) = \sum_x p_{X,Y}(x,y).$$

Independence

X and Y are independent if

$$p_{X,Y}(x,y) = p_X(x)p_Y(y) \quad \text{for all } x, y.$$

Conditional PMF

Definition

If $p_Y(y) > 0$, the conditional PMF of X given $Y = y$ is

$$p_{X|Y}(x|y) = \mathbb{P}(X = x|Y = y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}.$$

Multiplication rule

$$p_{X,Y}(x, y) = p_Y(y)p_{X|Y}(x|y).$$

Total Probability and Total Expectation

Total probability for PMFs

$$p_X(x) = \sum_y p_Y(y) p_{X|Y}(x|y).$$

Total expectation theorem

$$\mathbb{E}[X] = \sum_y p_Y(y) \mathbb{E}[X|Y = y].$$

Meaning

Average over conditional averages.

Worked Example 1: Defective Components

A box has 10 components, of which 3 are defective. Two components are selected without replacement. Let

$$X = \text{number of defective components selected.}$$

Find the PMF of X.

Possible values

$$X \in \{0, 1, 2\}.$$

Solution: Defective Components PMF

Use combinations because order does not matter.

$$\mathbb{P}(X = k) = \frac{\binom{3}{k}\binom{7}{2-k}}{\binom{10}{2}}, \quad k = 0, 1, 2.$$

k	Calculation	$\mathbb{P}(X = k)$
0	$\binom{3}{0}\binom{7}{2}/\binom{10}{2}$	$21/45$
1	$\binom{3}{1}\binom{7}{1}/\binom{10}{2}$	$21/45$
2	$\binom{3}{2}\binom{7}{0}/\binom{10}{2}$	$3/45$

Check:

$$\frac{21}{45} + \frac{21}{45} + \frac{3}{45} = 1.$$

Worked Example 2: Lottery Tickets

A small lottery sells 100 tickets. One ticket wins P500, four tickets win P100, and the rest win nothing. You buy one ticket. Let $X =$ amount won.

PMF

$$\mathbb{P}(X = 500) = \frac{1}{100}, \quad \mathbb{P}(X = 100) = \frac{4}{100}, \quad \mathbb{P}(X = 0) = \frac{95}{100}.$$

Find $\mathbb{E}[X]$.

Solution: Lottery Expected Value

$$\mathbb{E}[X] = \sum_x x p_X(x).$$

Substitute:

$$\mathbb{E}[X] = 500\left(\frac{1}{100}\right) + 100\left(\frac{4}{100}\right) + 0\left(\frac{95}{100}\right).$$

$$\mathbb{E}[X] = 5 + 4 + 0 = 9.$$

Interpretation

The expected winning per ticket is P9. If a ticket costs more than P9, the expected net gain is negative.

Worked Example 3: Server Requests

The number of requests arriving at a server in one minute has PMF:

x	0	1	2	3
$p_X(x)$	0.10	0.30	0.40	0.20

1. Find $\mathbb{P}(X \geq 2)$.
2. Find $\mathbb{E}[X]$.
3. Find $\text{Var}(X)$.

Solution: Server Requests

$$\mathbb{P}(X \geq 2) = \mathbb{P}(X = 2) + \mathbb{P}(X = 3) = 0.40 + 0.20 = 0.60.$$

$$\mathbb{E}[X] = 0(0.10) + 1(0.30) + 2(0.40) + 3(0.20) = 1.70.$$

$$\mathbb{E}[X^2] = 0^2(0.10) + 1^2(0.30) + 2^2(0.40) + 3^2(0.20) = 3.70.$$

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = 3.70 - (1.70)^2 = 0.81.$$

Practice Questions

1. A fair coin is tossed 4 times. Let X be the number of heads. Find the PMF of X.
2. A fair die is rolled once. Let $X = 1$ if the result is even, and $X = 0$ otherwise. Identify the distribution of X and find $\mathbb{E}[X]$ and $\text{Var}(X)$.
3. A random variable has PMF $p(0) = c$, $p(1) = 2c$, $p(2) = 3c$. Find c, $\mathbb{E}[X]$, and $\text{Var}(X)$.
4. In a batch of 20 laptops, 5 are defective. If 3 are selected without replacement, find the probability that exactly one is defective.

Solutions to Practice Questions

1. $\mathbb{P}(X = k) = \binom{4}{k}(1/2)^4 = \binom{4}{k}/16,\ k = 0, 1, 2, 3, 4.$
2. $X \sim \text{Bernoulli}(1/2),\ \mathbb{E}[X] = 1/2,\ \text{Var}(X) = 1/4.$
3. $c + 2c + 3c = 1 \Rightarrow c = 1/6.$ Thus

$$\mathbb{E}[X] = 0 \cdot \frac{1}{6} + 1 \cdot \frac{2}{6} + 2 \cdot \frac{3}{6} = \frac{4}{3}.$$

4. $$\mathbb{P}(X = 1) = \frac{\binom{5}{1}\binom{15}{2}}{\binom{20}{3}} = \frac{525}{1140} \approx 0.4605.$$

Summary

- A random variable maps outcomes to real numbers.
- A discrete random variable takes values in a finite or countable set.
- A PMF assigns probability to each possible value and must sum to 1.
- Expectation is a weighted average: $\mathbb{E}[X] = \sum_x x p_X(x)$.
- Variance measures spread: $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$.
- Common discrete distributions include Bernoulli, uniform, binomial and geometric.

End of Lecture

Questions?